MIL-M-43576

29 March 1968

MILITARY SPECIFICATION

MANTLE, ILLUMINATING (GASOLINE LANTERN)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and one grade of a knitted, soft-type mantle for use as an essential component part of a gasoline pressure lantern (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-T-191 - Textile Test Methods
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Fiberboard
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-L-1594 - Lantern, Gasoline, Leaded or Nonleaded Fuel, with Case

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6260

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3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, sample mantles covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all the applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.- The mantle (see figure 1) shall be made from a chemically treated seamless tubing knitted of rayon yarn. The bottom or closed end of the mantle shall be formed by gathering or pleating the tube and sewing with chemically treated cotton thread. The resulting bag shall then be turned so that the stub end is on the inside. The top section of the mantle shall be treated with a chemical solution to harden and increase the strength and abrasion resistance of this section of the mantle after ignition. Approximately half of this treated band shall be turned in, to form a double thickness. An asbestos drawstring shall be threaded around and through the top or head hardened section of the mantle for fastening the mantle to the mantle holder of the lantern.

3.2.1 Seamless knitted tubing.- The seamless tubing used in the manufacture of the mantle shall be knit from 150 denier, 40 filament, bright rayon yarn using a "duplex weave" or stitch.

3.2.2 Gathering or pleating thread.- The thread used to close the bottom of the tube of the mantle shall be a white chemically treated cotton thread.

3.2.3 Head hardening treatment.- The top 1-1/4 plus 3/16, minus 1/8 inches of the mantle shall be treated with a chemical solution to increase the after ignition strength of this section of the mantle. The hardening solution shall contain a dye which would show the extent of treatment.

3.2.4 Asbestos drawstring.- The drawstring shall be a 4 ply, 35 cut asbestos thread, 7, plus 1, minus 1/2 inches in length.

3.3 Size.- The mantle, as received and prior to ignition, shall conform to the dimensional requirements specified in figure 1. The dimensions are based on the flat configuration of the mantle.

3.4 Performance.- When tested as specified in 4.2.3, after ignition, the mantle shall assume an overall symmetrical shape and retain this shape throughout the test. At the conclusion of the test, the mantle shall measure at least 1-1/2 inches in length. Not more than one of ten mantles tested shall show more than one broken thread to produce a hole greater than 1/4 inch in diameter.

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3.4.1 Light output.- When tested as specified in 4.2.3, the mantle shall have a light output of not less than 120 candle power.

3.5 Weight of ash in mantle.- When tested as specified in 4.2.3, the weight of ash in the mantle (with drawstring removed) shall be not less than 0.320 grams.

3.6 Workmanship.- The mantle shall be free of tears, rips, and holes that may affect the performance of the mantle.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects specified in 4.2.2.2, dimensions specified in 4.2.2.3, and tested as specified in 4.3.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling an inspection lot for examinations and tests shall consist of all the mantles submitted for inspection at one time.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents. In addition, the supplier shall submit to the contracting officer or his authorized representative a certificate of compliance covering the material requirements contained in paragraphs 3.2.1 through 3.2.4 inclusive for every lot of component materials. The certificate shall be accompanied by actual test, inspection or other verifiable quality data.

4.2.2 Inspection of the end item.-

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4.2.2.1 Examination of the end item.- The end item shall be examined for defects in accordance with applicable subparagraphs and at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.5. Each lot of end items shall be examined for visual, dimensional, and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in number of mantles for the examinations in 4.2.2.2 and 4.2.2.3, and in units of shipping containers for the examination in 4.2.2.4.

4.2.2.2 Examination for visual defects.- The sample unit for this examination shall be one mantle.

Examine	Defect
Mantle	Bottom of mantle unraveled.
	Stub end on the outside.
	Top of mantle not dyed.
	Treated band not doubled over.
	Drawstring missing; not threaded around and through the top.
	Drawstring less than 6-1/2 inches.
	Drawstring more than 8 inches.
	Any tear, rip, or hole that may affect the performance of mantle.

4.2.2.3 Examination for dimensional defects.- The sample unit for this examination shall be one mantle.

Examine	Defect
Mantle	
Width	Less than 2-1/8 inches.
	More than 2-3/8 inches.
Length (head hardened section turned in)	Less than 2-5/8 inches.
	More than 3-1/8 inches.
Head hardened area (full length, see 3.1.3)	Less than 1-1/8 inches.
	More than 1-7/16 inches.
Head hardened area (doubled over)	Less than 5/8 inch.
	More than 7/8 inch.

4.2.2.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping container shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

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Examine	Defect
Material	Any nonconforming component; component missing or damaged. Inadequate application of components, such as: incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling .
Contents	Envelope not fabricated from material specified. Envelope not securely closed. Mantle missing. Number of unit packages per intermediate package or packages per shipping container is more or less than the specified or indicated quantity. 1/ Bulged or distorted container.
Markings, (exterior and interior)	Omitted, incorrect, incomplete, not in accordance with contract requirements.

1/ No more than five intermediate packages from a shipping container shall be taken for the unit packaging examination.

4.2.2.5 Inspection levels and acceptable quality levels (AQLs) for examination.- The inspection levels for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per one hundred units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.2.2.2	S-4	2.5
4.2.2.3	S-2	2.5
4.2.2.4	S-2	2.5

4.2.3 Testing of the end item.- The mantle shall be tested as specified in table I for each inspection lot. The sample unit for testing shall be 12 mantles. The lot size, for purposes of determining the number of sample units to be tested, shall be expressed in units of intermediate containers (six mantles) and the number of sample units shall be determined as indicated below. Each sample unit shall be drawn from different shipping containers when practicable. The lot shall be unacceptable if one or more of the sample units fail to meet the test requirements specified.

Lot size (No. of intermediate containers)	No. of sample units
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations per sample unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	1/ Numerically to Nearest		
Performance	3.4	4.3.1.2	X		10	X			
Light output	3.4.1	4.3.1.3	X		1	X			
Weight of ash in mantle	3.5	4.3.2	X		2		Average of two determinations to nearest 0.001 gram		

1/ If failure is indicated, report either description of failure or the numerical point of failure, as applicable

AMXRE Form 598
8 February 1965

MIL-M-43576

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4.3 Test procedures.-

4.3.1 Performance and light output tests.-

4.3.1.1 Equipment.-

4.3.1.1.1 Lantern.- A single-mantle gasoline lantern conforming to MIL-L-1594, with a pressure guage installed, shall be used. Unleaded gasoline shall be used as fuel during the tests.

4.3.1.1.2 Light meter.- A foot candle meter equipped with Viscor and cosine correction filters shall be used (Weston Model 614 or equal). The meter shall be fitted with a 4-inch diameter tube, lined with black felt, in front of the cell. The length of the tube, when placed against the lantern globe, shall space the face of the cell a distance of 17 inches (1.414 feet or the $\sqrt{2}$ feet) from the center of the mantle.

4.3.1.2 Performance test.- Install the test mantle in the lantern and light in accordance with the instruction specified in MIL-L-1594, with an initial air pressure of 20-22 pounds per square inch (psi). Operate the lantern for 30 minutes, turn off, and cool for 30 minutes. Repeat this operation for a total of five times. During the test, observe the mantle for changes in appearance or shape and, at the conclusion of the test, measure the length of the mantle. Use this mantle for light output measurement. Install and ignite the nine remaining mantles, igniting each mantle only once. Observe broken threads and measure the size of any hole.

4.3.1.3 Light output.- Using the multi-ignited mantle as tested in 4.3.1.2, light the lantern and operate in accordance with instructions above, maintaining an air pressure of 20-22 psi. Place the lighted lantern in a normally illuminated room, excluding any other intense light source, natural or artificial, and measure the light output of the mantle using the foot candle meter, fitted with tube described in 4.3.1.1.2. Calculate the light output in candle power as follows:

$$\text{Candle power} = \text{foot candles} \times D^2$$
$$= \text{foot candles} \times 2$$

where:

D_2 = distance (in feet) between center of mantle and sensing cell of meter.

$D_2 = 2$, when D = 1.414 ft. (17 inches).

4.3.2 Weight of ash in mantle.- The weight of ash in mantle shall be determined as specified in paragraph 5.7 of Method 2611 in CCC-T-191.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.-

5.1.1.1 Unit packaging.- Each mantle shall be placed in a close-fitting coin-type envelope fabricated of kraft or glassine paper, or polyethylene film bag. The envelope flap shall be closed and sealed. The polyethlene bag closure shall be heat sealed.

5.1.1.2 Intermediate packaging.- Six mantles, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting paperboard box conforming to variety plain, style II, type A, class a of PPP-B-566. Each box shall be closed in accordance with the appendix of PPP-B-566.

5.1.2 Level C.- Mantles shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- One thousand seven hundred and twenty-eight mantles, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A packages shall be packed flat 6 in length, 4 in width and 12 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 26 inches in length, 13 inches in width and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- One thousand seven hundred and twenty-eight mantles, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat 6 in length, 4 in width and 12 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 26 inches in length, 13 inches in width and 9-3/4 inches in depth. Approximate dimensions are furnished as a quide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Mantles, packaged as specified in 5.1, shall be packed in a manner to issure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

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5.3 Marking.- In addition to any special marking required by the contract or order, unit, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The mantle covered by this specification is intended for use in a single-mantle gasoline lantern which conforms to the requirements of MIL-L-1594; Lantern, Gasoline, Leaded or Nonleaded Fuel, with Case.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

Custodians:

Army - GL
Air Force - 82

Review activities:

Army - MD
Air Force - 82, 85

User activities:

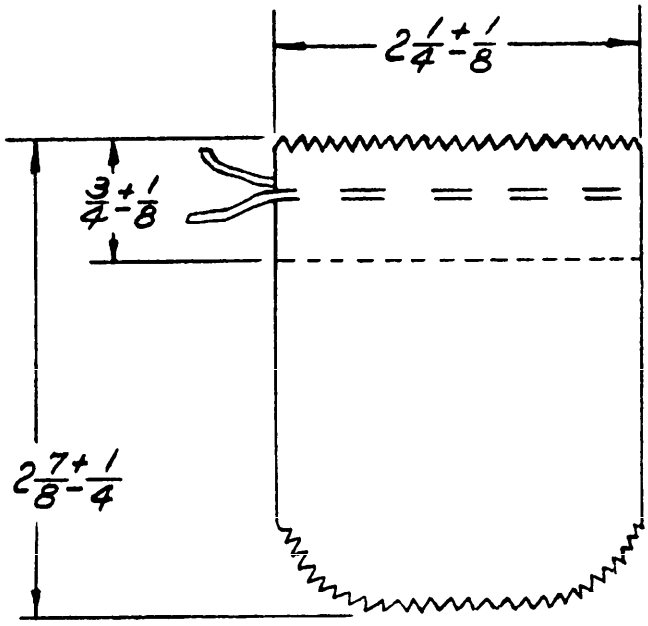
Army - ME
Navy - CG, MC

Preparing activity:

Army - GL

Project No. 6260-0028

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WITH HEAD HARDENED
SECTION TURNED IN

FIG. 1 MANTLE, ILLUMINATING (GASOLINE LANTERN)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Mantle, Illuminating (Gasoline Latern) MIL-[illegible]-43576

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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